



Business Intelligence

D7044E

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Case Study (A) + Project

0002	Individual task	1.5	TG U G#
0003	Group/Project work	2	TG U G#

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Sports Analytics – Football Analytics*

The application of analytics to business problems is a key skill. Many of these techniques are now being used to make decisions in all aspects of sports, a very hot area called *sports analytics*. *Sports analytics* is the art and science of gathering data about players and teams to create insights that improve *sports decisions*, such as:

- *Which players to recruit?*
- *How much to pay a player?*
- *Who to play?*
- *How to train players?*
- *How to keep players healthy?*
- *When to retire a player?*

For *teams*, it also involves other business *decisions*, such as:

- *How much to price a ticket?*
- *How to analyze competitors' strength and weaknesses?*

Indeed, *sports analytics* is becoming a specialty within analytics. It is an important area since sports is a big industry.

The use of data and analytics has the potential to turn a losing team into a winning team. Now, analytics are being used in various aspects of sports. *Sports analytics* can be divided into: *Front Office* and *Back Office*.

Front Office analytics include fans behavior such as season tickets renewal and regular ticket sales. On the other hand, *Back Office* focuses on the analysis of both individual players as well as team play. For *individual players*, there is a focus on recruitment models, analytics for strength and fitness and analytics for avoiding overtraining and injuries. *Team Analytics* include strategies and tactics, competitive assessments, optimal team formation, and during the match decisions.

The following use case scenarios illustrate potentials of football analytics to improve the game, in the same way analytics have improved traditional industry's decision making.

* Source [**Adapted**]: Sharda, Delen, and Turban, 2018, Business Intelligence, Analytics, and Data Science, Pearson, pp. 4-10.

Scenario 1: Business Development

David Ward works as a *business analyst* for a major *football* team, focusing on revenue. He analyzes ticket sales. Both from season ticket holders as well as single-ticket buyers' perspectives. Sample questions in his area of responsibility include:

- *Why season tickets (not) renew?*
- *What factors drive last-minute individual single ticket purchases?*
- *How to price the ticket?*

David uses simple statistics on fan behavior like overall attendances and answers to survey questions about likelihood to purchase again. However, what fans say versus what they do may differ. David runs a survey on fans by ticket seat location ("Tier") and asks about their likelihood of renewing their season tickets. But when he compares what they say versus what they do, he discovers big differences. See below Figure (1).

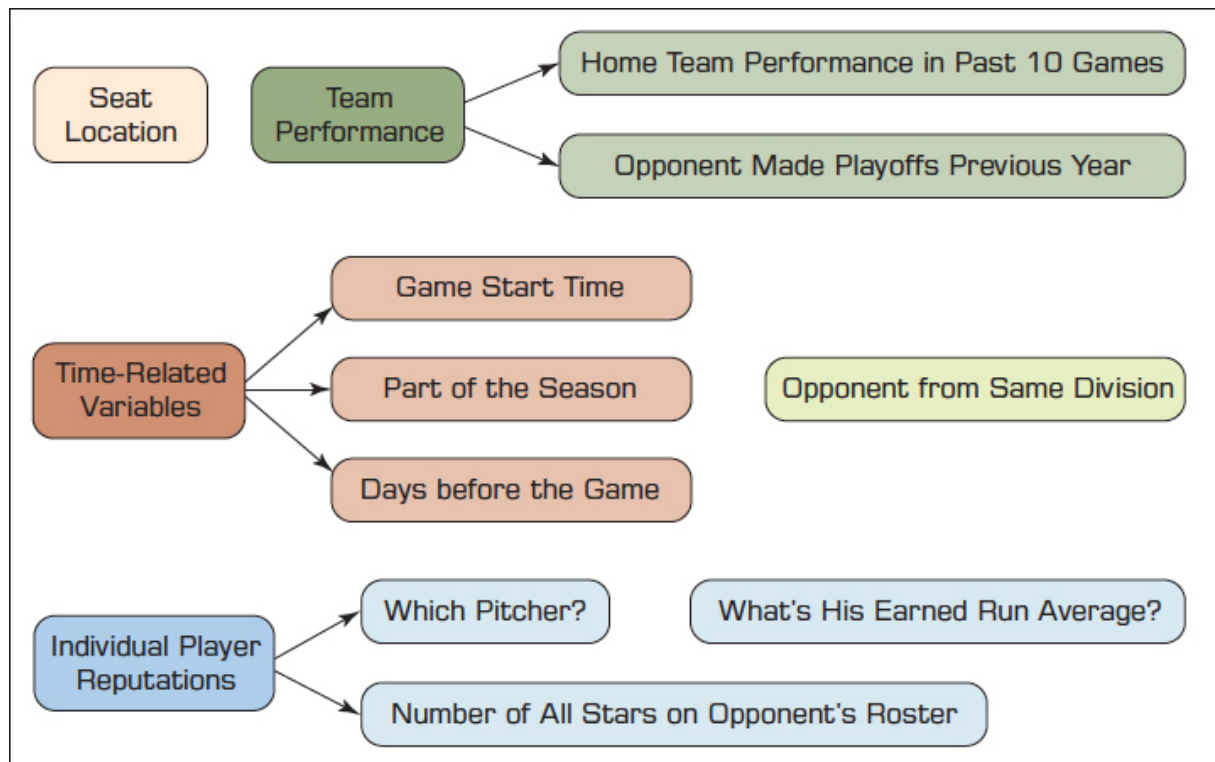
He found that 69% of fans in Tier 1 seats who said on the survey that they would "Probably Not" Renew, actually did. This is useful insight that leads to actions – customers in the green cells are the most likely to renew tickets, so require fewer marketing budget to convert, for example, compare to customer in the blue cells.

Tier	Highly Likely	Likely	Maybe	Probably Not	Certainly Not
1	92	88	75	69	45
2	88	81	70	65	38
3	80	76	68	55	36
4	77	72	65	45	25
5	75	70	60	35	25

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Figure (1): Season Ticket Renewals – Survey Scores

However, many factors influence fan ticket purchase behavior, specially price, which drives more sophisticated statistics and data analysis. For both areas, but especially single-game tickets, David is driving dynamic pricing moving the business from simple static pricing by seat location tier to day-by-day up-and-down pricing of individual seats. This is a huge research area, and huge upside potential for revenue enhancement. For example, David's pricing takes into account the team's record, who they are playing, game dates and times, which starts play for each team, fan buying history of both single and season tickets, seat location, number of seats, and other factors such as traffic congestion during game time and the weather. See below Figure (2).



Source: Adapted from C. Kemper and C. Breuer, "How Efficient is Dynamic Pricing for Sports Events? Designing a Dynamic Pricing Model for Bayern Munich", *Intl. Journal of Sports Finance*, 11, pp. 4-25, 2016.

Figure (2): Factors influencing Dynamic Pricing

Which of these factors are important? How much? Given the extensive statistics background, David builds different models – e.g., Regression - to pick out key factors driving these historic behavior. Also, uses clustering to segment fans. Added, interesting in running trend lines to forecast season sales.

Scenario 2: The Coach

Bob Lewis is the football coach of a team. For him, it is all about winning the games. His areas of focus include recruiting the best players, developing the team offense and defense skills, and getting the maximum efforts from the players on the day of the game. Sample questions in his area of responsibility include: *Who do we recruit? How hard to push the players? Where are opponents strong or weak at? What training/training tools best develop players skills?*

Fortunately, his team has hired a new *Assistant*, Anna Eriksson, who specializes in helping the coaches make tactical decisions. She is working to create own and opponents analytics. Her dream is to have a *dashboard* of multiple analytics on own team and another one on each opponent. They two need to merge and being presented to the manager before each and every game.

Note:

- *Please refer to the course material on Tableau!*
- *Added, refer to material on chapters: 1,2, & 6 in order to think in line with Anna!*

Scenario 3: Team Doctor – AKA Medical Director

Dr. Jan Johnson is the football team Doctor. His job is to help the players stay healthy and to advise the coaches on how much load to put on players during practices. He also has an interest in players well-being, including how much they sleep and how much rest they get between heavy and light practice sessions. The goal is to ensure that the players are ready to play on game days at maximum efficiency.

Fortunately, because of wearables, there is much more data for his to analyze. His players train using vests that contain sensors that can measure internal loads like heartbeats, body temperatures, and respiration rates. The vests also include accelerometers that measure external loads like running distance, and speeds as well as accelerations and decelerations. He knows which players are giving maximal effort during practices and those who are not.

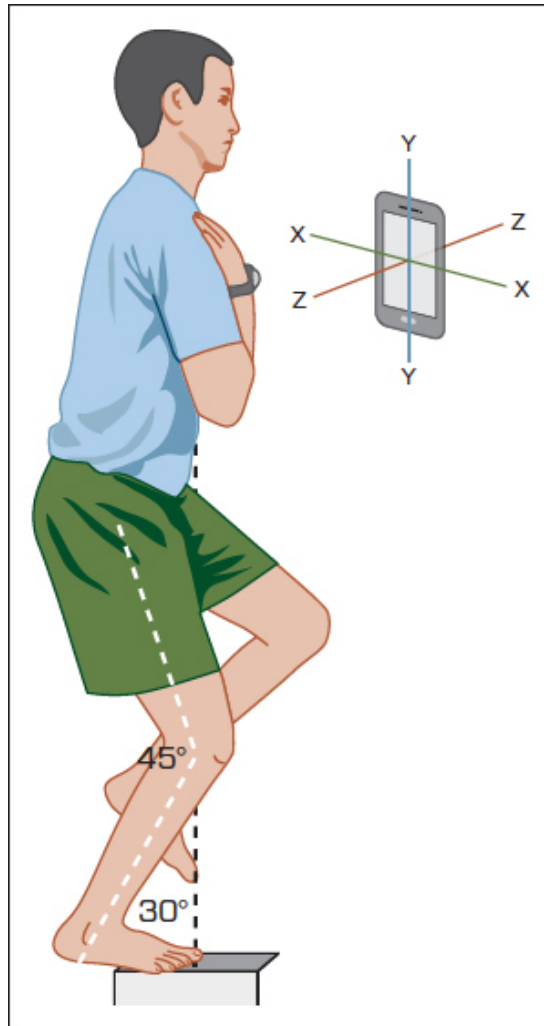
His focus at the moment is research that predicts or prevents player injuries. See below figure (3).



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Figure (3): Injury Models

Some simple tasks like a single leg Squat Hold Test, standing on one foot, then the other, with score differentials of more than 10% can provide useful insights on body core strength and weaknesses. See below figure (4).



(Source: Figure adapted from Gary Wilkerson and Ashish Gupta).

Figure (4): Core Body Strength Test

If a player is hit hard during a match, a trainer can conduct a sideline test, reacting to a stimulus on a mobile device, which adds to traditional concussion protocols. Sleep sensors show who is getting adequate rest (or who partied all night). He has the MRI lab on campus to do periodic brain scans to show which athletes are at risk for brain injury.

Case Questions, 1.5 credits [Individual]

- 1) Using a table format, list 10 decisions that could be supported with football analytics:
 - For each, define who the decision maker is
 - Classify the decision by managerial level, i.e., operational, tactical, strategic
 - Classify the decision also by complexity, i.e., structured, semi-structured, unstructured
 - Identify variables: predictor/independent and response/dependent
 - Recommend a model or tool to support the decision
- 2) How does analytics influence, positively or negatively, the good of the game?
- 3) Can football analytics be generalized to other sports? How?

Submission [Individual]:

- Individual piece of unaided work
- Preferably, a maximum 5 A4 pages
 - *Hint: please see Appendix!*
- Time New Roman font 12, single-spaced, formatted “justify”
- Put your references in the end
- File format: **.PDF** only, student name is the prefix
- No submission via email!

Project Requirements, 2 credits [Group]

- 1) In Tableau:
 - 8 sheets: 4 for the team & 4 for the opponents
 - 2 dashboards: 1 for the team & 1 for the opponent
 - 1 story; of the 2 dashboards
- 2) In a .PDF file:
 - Project explanation including data, sheets, objectives, pre-processing, etc.

Assumptions:

- Use your Tableau-earned skills during the course
- Fetch your dataset
- It is up to you which league or football team to focus your analysis at
 - Flexibility: feel free to choose any sport, should that help to make a better project!

Submission [Group]:

- Group unaided work
- File format:
 - Tableau **.twbx** File & **.PDF** explanatory and stepwise file, group name is the pre-fix
 - Direct **Link** to the data which has been used by the group, via dropbox or OneDrive
- No submission via email!

Resources:

- <https://public.tableau.com/en-us/s/gallery/?tab=viz-of-the-day&type=viz-of-the-day>
- <http://visualoop.com/blog/3199/25-great-data-visualizations-made-with-tableau-public>
- <https://www.tableau.com/solutions/gallery>
- <https://www.data.gov>
- <https://data.europa.eu/euodp/en/data>
- <https://www.kaggle.com/datasets>
- <https://www.scisports.com/>
- <http://analyticsfc.co.uk/>
- <https://www.optasports.com/sports/football/>
- https://www.footballbenchmark.com/data_analytics
- <https://www.whoscored.com/Statistics>
- <https://www.beinsports.com/en/football/statistics>
- <https://www.stats.com/data-feeds/>

Appendix

Individual Assignment - Cheat Sheet

Motivation:

In the below table, I will focus on 10 different football analytics decisions. The reason I have selected those decision are:

- 1..
- 2...
- 3..

There exist a list of models and tools. The combination which I have proposed is attributable to the below reasons:

- 1...
- 2...
- 3...

Decision	Decision Maker	MGT. Level	Decision Type	Variables	Model/Tool
D-1: how to increase revenue from single ticket purchase	Business Analyst	Operational	Semi-structured	IND: ticket price, promotion, seat availability, match datetime DEP: ticket purchase	Linear Regression
D-2: predict player injury	Team Doctor	Operational	Semi-structured	IND: sensor readings, amount of exercise, sleep, previous injury, DEP: player injury	Decision trees
D-3:					
D-4:					
D-5:					
D-6:					
D-7:					
D-8:					
D-9:					
D-10:					